CO1-AT2 SHORT ANSWERS

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Q1. 45 MHz ÷ 25 kHz = 1800 simplex channels. Full-duplex channels = 1800 ÷ 2 = **900 channels**.

Q2. Channels per cell = 560 ÷ 7 = **80 channels**.

Q3. Total voice channels = 100 × 8 = **800 simultaneous voice channels**.

Q4. Power gap = 10 log■■(80/4) = **13.01 dB**.

Q5. C/I = –60 – (–92) = **32 dB**.

Q6. Total channels = 42 MHz ÷ 300 kHz = 140. Per cell = 140 ÷ 7 = **20 channels**.

Q7. With n = 4, doubling distance reduces power by 2■ = 16. New power = 64/16 = **4 W**.

Q8. For inverse-square law, $P \propto R^2$. Doubling radius requires 2^2 = **4× transmit power**.

Q9. Total cluster channels = 4 × 15 = **60 channels**.

Q10. f■ = vf/c = (100/3.6 × 900×10■)/(3×10■) ≈ **83.3 Hz**.

Q11. Slot duration = 5 ms ÷ 8 = **0.625 ms** = 625 μs.

Q12. Carrier frequencies = 24 MHz ÷ 0.2 MHz = **120 carriers**.

Q13. 8 W = 39.03 dBm. Received power = 39.03 – 33 = **6.03 dBm**.

Q14. FSPL = 20 log■■(4πd/λ), for 1900 MHz and 2.5 km: ≈ **106 dB**.

Q15. Cluster capacity = 6 × 250 = **1500 simultaneous calls**.

Q16. Active calls = 4000 calls/s × 180 s = **720,000 calls**.

Q17. Bandwidth = 150 × 25 kHz = **3.75 MHz**.

Q18. There are 29 gaps between 30 bands. Wasted bandwidth = 29 × 40 kHz = **1.16 MHz**.

Q19. Speed = 90 km/h = 1.5 km/min. Cell diameter = 3 km. Handoffs/hour = 60/2 = **30 handoffs**.

Q20. Total users = 8 × 40 × 250 = **80,000 users**.

Q21. Cell area ≈ $R^2 = 0.4^2 = 0.16$ km². Number of cells = 400/0.16 = **2500 cells**.

Q22. Channels = 6 MHz ÷ 30 kHz = **200 channels**.

Q23. Traffic = 1200 × 4 × (2.5/60) = **200 Erlangs**.

Q24. Maximum users = 40 MHz ÷ 400 kHz = **100 users**.

Q25. Total capacity = 1200 × 600 = **720,000 users**.

Q26. Paging rounds = 200 ÷ 4 = **50 rounds**.

Q27. Offered traffic = 40,000 × 3 = 120,000 Erlangs. Blocked fraction = (120,000–100,000)/120,000 = **16.67%**.

Q28. Roaming subscribers = 15% × 600,000 = **90,000 subscribers**.

Q29. Utilization = (9000/12000) × 100 = **75%**.

Q30. Total channels = 48 MHz ÷ 200 kHz = 240. Per cell = 240 ÷ 4 = **60 channels**.

Q31. Channels = 30 MHz ÷ 200 kHz = **150 channels**.

Q32. Successful handoffs = 2500 × 0.99 = **2475 handoffs**.

Q33. 12 W = 40.79 dBm. Received power = 40.79 – 45 = –**4.21 dBm**.

Q34. Cluster capacity = 7 × 900 = **6300 users**.

Q35. Slot duration = 4.8 ms ÷ 8 = **0.6 ms** = 600 μs.

Q36. Doppler shift = (150/3.6 × 1.8×10■)/(3×10■) = **250 Hz**.

Q37. Offered traffic = 70,000 × 4 = 280,000 Erlangs. Blocked share = (280,000–250,000)/280,000 = **10.71%**.

Q38. Cluster capacity = 9 × 600 = **5400 users**.

Q39. Dropped calls/s = 60,000 × 0.004 = **240 calls/s**.

Q40. Path loss = 43 – (–87) = **130 dB**.

Q41. Total area = 150 × 2.5 = **375 km²**.

Q42. Total channels = 48 MHz ÷ 0.2 MHz = 240. Per cell = 240 ÷ 6 = **40 channels**.

Q43. Paging cycles = 150,000 ÷ 6000 = **25 cycles**.

Q44. Handoffs = 9 ÷ 3 = **3 handoffs**.

Q45. SNR = –85 – (–115) = **30 dB**.

Q46. Required data rate = 250 × 150 kbps = **37.5 Mbps**. Assuming 1 bps/Hz, minimum bandwidth ≈ **37.5 MHz**.

Q47. Traffic = 1500 × 4 × (2.5/60) = **250 Erlangs**.

Q48. Control-channel bandwidth = 250 × 12 kHz = **3 MHz**.

Q49. Total throughput = 1500 × 600 kbps = **900 Mbps**.

Q50. Tower area = $\pi(4^2)$ = 50.27 km². Towers = 350/50.27 ≈ 6.96 → **7 towers**.